Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/24/2005
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Project Title:	Convective and Morphological Instabilities During Crystal Growth		
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Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
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Task Description:	This research consists of theoretical studies of the interaction of fluid flow with the crystal-melt interface. When growth is by step motion, kinetic anisotropy provides enhanced morphological stability. Shear flows along the interface interact strongly with the step motion and cause decreased stability for a flow in the same direction as the step motion and enhanced stability for flows counter to step motion. We plan to investigate these flow-interface interactions for a wide range of processing conditions for melt and solution growth with extension to more complex physical models and nonlinear interface morphologies. Calculations of solute segregation and interface stability for specific alloys and processing conditions relevant to past and future space experiments will also be performed. During the directional solidification of a binary alloy solute inhomogeneities can arise from both fluid flow and morphological instability. In microgravity buoyancy-driven fluid flow is reduced, and experiments to study the evolution of morphological patterns without the interference of fluid flow may be possible. Another goal is to develop an understanding of the interaction of		

fluid flow with the crystal-melt interface so that solute segregation in solidifying alloys can be controlled in order to obtain materials with superior properties on Earth as well as in a microgravity environment. Solute segregation during alloy solidification and crystal growth determines the properties of the solidified material. The length scale associated with the spatial distribution of solute (microsegregation) is established by the morphology of the crystal-melt interface (cellular or dendritic). Fluid flow in the liquid results in macrosegregation of solute. This research will determine the processing conditions required to eliminate microsegregation and produce alloys with desirable microstructures and improved properties.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Since the original paper by Jackson and Hunt on lamellar and rod eutectic growth, there have been numerous papers that have extended and generalized their analysis. The analysis has also been applied to the solidification of monotectic alloys. The original Jackson-Hunt analysis neglected fluid flow in the liquid phase; fluid flow can occur due to buoyancy effects, density differences between the phases, forced convection, and, in the case of monotectics, Marangoni flows. One of the basic assumptions of the Jackson-Hunt analysis is that the concentration field in the melt phase can be calculated approximately by assuming that the melt-eutectic interfaces are planar; the validity of this assumption has been investigated numerically by a number of studies using boundary integral techniques.
During the project the investigators extended the Jackson-Hunt model of eutectic growth to allow for different densities of the phases. The density differences give rise to fluid flow that is calculated from a series solution of the fluid flow equations in the Stokes flow approximation. The solute diffusion equation with flow terms is then solved numerically using an adaptive refinement and multigrid algorithm (PLTMG). The interface undercoolings and volume fractions are calculated as a function of spacing for tin-lead and iron-carbon eutectic alloys and for an aluminum-indium monotectic alloy. The numerical results have been compared with various approximations based on the Jackson-Hunt analysis.

In a collaborative effort with S. Van Vaerenbergh, Universite Libre de Bruxelles, the effect of a temperature-dependent solute diffusion coefficient on the morphological stability of a binary alloy during directional solidification has been treated by a linear stability analysis. The Soret effect (thermodiffusion) is also included in the analysis. Specific calculations have been carried out for a tin alloy containing silver for which the diffusion coefficient has a linear dependence on temperature. Although the temperature dependence of the diffusion coefficient has little effect on the critical concentration for the onset of morphological stability, it causes a significant change in the wavelength at the onset of instability. Publications in this area have discussed our recent research on temperature-dependent solute diffusivity and electrical effects during growth of a binary alloy from the melt and on anisotropic kinetics and shear flows during growth from a supersaturated solution. For these cases, we computed the stability of a planar interface during constant velocity growth. We also discussed the effect of a fluid flow due to density change upon monotectic solidification. The separation of scales during dendritic growth of a binary alloy due to the large ratio of thermal and solutal diffusivity was reported as well.

We have formulated the linear stability problem for directional solidification of a two-phase or monotectic system, focussing on the behavior of a liquid-liquid interface that is subject to morphological and convective instabilities. This work should help explain the effect of convection on rod and lamellar spacing during cooperative growth in such systems, including the specific flow mechanisms that affect the observed length scales. We have also considered the effect of surface tension anisotropy on the classical Rayleigh instability of cylindrical interfaces with uniform cross-sections. The Rayleigh instability arises in a number of materials processing applications, particularly at small length scales where surface energy has a strong effect on interface morphology. For crystalline interfaces, the underlying anisotropy of the surface energy is found to have a significant affect on the stability of the interface to axial pertubations that ultimately lead to break-up of the interface.

Research has included the consideration of the effect of fluid flow upon monotectic solidification. This work is intended to help explain the effect of convection on rod and lamellar spacing during cooperative growth in such systems, including the specific flow mechanisms that affect the observed length scales. We have formulated the linear stability problem for directional solidification of a two-phase or monotectic system, focusing on the behavior of a liquid-liquid interface that is subject to morphological and convective instabilities. We have considered flows generated by buoyancy-induced convection, density-driven convection, Marangoni convection, and pressure-driven convection associated with the Laplace-Young boundary condition at the fluid-fluid interface. We are currently examining the case of Marangoni convection in some detail, in order to highlight the differences between such convection in inert layers of immiscible fluids and in two-phase systems where phase transformations may occur. A manuscript describing this work is currently in preparation.

We have also continued our studies on the effect of surface tension anisotropy on the classical Rayleigh instability of cylindrical interfaces with uniform cross-sections. The Rayleigh instability arises in a number of materials processing applications, particularly at small length scales where surface energy has a strong effect on interface morphology. For crystalline interfaces, the underlying anisotropy of the surface energy is found to have a significant affect on the stability of the interface to axial perturbations that ultimately lead to break-up of the interface. Our work on the case of an isolated bridge configuration appeared in the Proceeding of the Royal Society. This work is being extended to include the effect of contact angles with an underlying substrate, such as occur during the processing of quantum wires; a manuscript describing this work is currently in preparation.

Another project concerns the stability of a crystal interface during electrodeposition. This is an extension of the classical Mullins-Sekerka analysis for morphological stability during directional solidification to the process of growth in the presence of an electric field. We consider the case of transport within the convective boundary layer, focusing on the effects of interfacial additives or catalysts that are found experimentally to stabilize the system. A publication describing this work appeared in the Journal of the Electrochemical Society, and another manuscript extending this work to the nonlinear regime has also appeared recently.

The recent interest in the design and operation of nanoscale systems has prompted increased attention to problems of stress and diffusion in thin layers and other small-scale geometries. The underlying process represents a complicated modeling problem in which the diffusion process is affected by the deformation due to plastic and elastic forces in the sample, and, conversely, the deformation field is affected by interdiffusion that may induce strain due to differences in the molar volumes of the various constituents. This inherent coupling makes an understanding of the process

challenging, and progress is often possible only through a combination of modeling, analysis, and computations. In this project the modeling of stress effects during interdiffusion of solute in a diffusion couple was considered. The work involves a continuum model that describes the creation and annihilation of lattice vacancies at defects in the crystal, which can produce swelling or shrinkage of the sample locally. The Kirkendall effect is a well-known consequence of a disparity in the intrinsic diffusion coefficients of the components in alloy diffusion couples. The effect has three manifestations: a shift of inert markers parallel to the diffusion direction (deformation with pure axial displacement), lateral shape changes (displacements with a component orthogonal to the diffusion direction), and the formation of voids. We have developed a model that includes a description of the uniaxial Kirkendall effect, capturing the deformation parallel to the diffusion direction, and at the same time obtaining the concommitant lateral deformation. A solution to the coupled equations describing diffusion and deformation can be obtained using Fourier analysis for the case under consideration. We obtain limiting behavior for thin and thick cylinders and slabs, and compute numerical results of the model for the lateral shape change that compare favorably with experimental measurments of the deformation observed in gold-silver alloys. This work is in press in Acta Materialia. An extension of this work is being used to model the bending of a thin plate, for which experimental data is also available. This work is intended to address problems in which stress is produced in nano-scale samples due to the effects of multicomponent diffusion.

Melting and solidification into either a crystalline or amorphous state in nano-volumes are basic mechanisms of modern phase change recording. This process exploits an amorphous-to-crystalline phase change to record information onto electronic media such as DVDs. Due to the density difference between phases, transformations in confined volumes are accompanied by internal stresses. Thus, it is necessary to take into account the effect of internal stress during phase change recording. In this project we have modeled this "writing" process by considering melting and solidification in a confined volume. To simulate melting/solidification we employ a phase field model. Phase-field modeling is an effective approach for the analysis and prediction of the evolution of complex microstructures during phase transformations. This approach has been particularly successful when it has been applied to solidification problems. For single-component system the standard phase-field method requires solving two equations: the time dependent Ginsburg-Landau phase-field equation and the equation of heat transfer. The phase field model that we develop includes stress field during non-isothermal solidification of a one-component system. These stresses can be external or internal, i.e. they can arise as a result of boundary conditions at the external surface or they can be a result of elastic interactions between the phases. Thus, the phase field model takes into account the simultaneous evolution of three fields: a phase field, a temperature field and a stress/strain field. To demonstrate this approach we consider the solidification and melting of a confined sphere. For this simple system the complete numerical solutions taking into account the time-space evolution of temperature, order parameter and strain-stress field is obtained. The simulation shows that at some boundary and initial conditions the evolution of a spherical symmetry system results in steady states corresponding time-independent distribution of order-parameter and uniform temperature. Comparison results of phase field modeling and thermodynamic analysis of a sharp-interface model shows that these steady states are equilibria that exist in systems with fixed volume. The analysis of these equilibrium states allows us to estimate the value of interface energy that is then verified by a phase field calculation of the energy of plane liquid/solid interface. The model can be used to simulate the process of "writing" to electronic media that exploit an amorphous-to-crystalline phase change for recording information.

The viscous fluid flow created between differentially rotating coaxial cylinders has provided a fertile testing ground for both linear and nonlinear stability theory. Beginning with the work of G.I. Taylor in 1923, numerous experimental and theoretical studies on flow transitions and morphologies of supercritical circular Couette flow have appeared. Initial studies on thermally-driven circular Couette flow were motivated by technological problems in the cooling of rotating electrical machinery. Early theoretical attacks neglected gravity and usually considered only axisymmetric disturbances in the limit of infinite aspect ratio. There has been a renewed interest in the problem of radially heated rotating flows, partially from a continued effort to enhance the cooling of rotating machinery, but also with the aim of understanding and controlling instabilities in nematic liquid crystal systems, in chemical vapor deposition processes, and in the solidification of pure metal and organic materials. The stability of viscous isothermal circular-Couette flow generated by rotation of the inner cylinder is controlled by the radius ratio, the aspect ratio, and the Taylor number. Stationary counter-rotating toroidal cells of uniform width stacked one above the other appear at a critical Taylor number. In the absence of rotation, natural convection between vertical differentially heated concentric cylinders depends crucially on the magnitude of the imposed thermal heating and the system aspect ratio. Early experimental studies fostered the identification of three distinct flow regimes in both planar and cylindrical gaps: conduction, transition, and convection. In this project, motivated in part by experimental work on instabilities of a cylindrical crystal-melt interface done by Glicksman et al. at RPI, we consider the convection regime, where axial boundary layers form along each cylinder walls, and a vertical temperature gradient develops in the interior of the sample. We have developed an analytical solution for the base flow in the convection regime in a vertical annulus with a rotating inner cylinder and a stationary outer cylinder. The stability of this base flow is then computed numerically with respect to both axisymmetric and asymmetric disturbances. This joint research with M. Ali (King Saud University, Riyadh, Saudi Arabia) has been submitted for publication in the Physics of Fluids. Related work on hydrodynamic instabilities has been performed in collaboration with D. Cotrell, a post-doctoral researcher at NIST who was previously supported by a NASA Graduate Student Researchers Program Fellowship at the University of Ilinois at Urbana-Champaign.

Bibliography Type:	Description: (Last Updated: 01/24/2005)
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Presentation	McFadden, G.B "Phase-Field Modeling of Electrochemistry" Department of Mathematical Sciences Department Seminar, New Jersey Institute of Technology, Newark, NJ Nov-2003
Presentation	McFadden, G.B "Interfacial boundary conditions and phase-field models of solidification, Nonequilibrium Interface Dynamics: Theory and Simulation from Atomistic to Continuum Scales" Center for Scientific Computation and Mathematical Modeling, University of Maryland, College Park, MD Oct-2003
Presentation	McFadden, G., Slutsker, J., Warren, J., Boettinger, W, Thornton, K., Roytburd, A., Voorhees, P "Surface Energy and Surface Stress in Phase-Field Models of Elasticity" 3rd Annual Workshop on the Evolution and Self-Assembly of Quantum Dots, Northwestern University Aug-2004